

Name: _____ Date: _____

Definition 1. Recall the limit definition of the derivative – if $f(x)$ is a (smooth) function, then the **derivative** of $f(x)$ is the function $f'(x)$ (read as f prime of x) which is defined as

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Problem 1. For each of the following functions, use the limit definition of the derivative to compute their derivatives. You will not receive credit for simply writing the answers, you **MUST** use the limit definition of the derivative to receive credit.

(a) $f(x) = x^2$

(b) $f(x) = x^3 - 3$

(c) $f(x) = 3x^2 + 2x - 1$

(d) $f(x) = \frac{1}{x}$

(e) $f(x) = \sqrt{x}$

(f) $f(x) = \sin(x)$. *Hint: the following “difference formula” may be helpful: $\sin(A - B) = \sin(A)\cos(B) - \cos(A)\sin(B)$. Also, at some point, looking back at our squeeze theorem worksheet may be helpful!*